

The Elevator Problem

Memo #1

From: Your Boss
To: You
Re: Late Arrivals

It has come to my attention that many of our employees are arriving in the offices after the 9:00 a.m. starting time. They complain that the four old elevators cannot handle the rush at the start of the day. We cannot, at this time, install new elevators or increase the speed or capacity of existing ones above the current ten persons.

Please investigate and let me have some possible solutions to the problem on Friday, September 6th.

Memo #2

From: You
To: Your Assistant
Re: Late Arrivals

Can you find out:

1. How long the elevators take to move between floors and how long they stop for people to exit?
2. How many people from each floor use the elevator in the morning?
3. How many people were late this morning?

Memo #3

From: Your Assistant

To: You

Re: Answers to your questions

1. The elevators appear to take 5 seconds between each floor, an extra 10 seconds for each stop, and another 8 seconds if the doors have to reopen. It also seems to take an average of about 15 seconds for the elevators to fill on the ground floor.
2. The number of workers on each floor are:

Floor	G	1st	2nd	3rd	4th	5th	6th
People	0	100	120	60	120	80	20

3. About 110 people clocked in after 9am this morning.

Memo #4

From: You

To: Your Boss

Re: Solutions to the lateness problem

Assignment: create a mathematical model that describes this situation quantitatively. Recommend reasonable measures for improving the situation, explain how these will be implemented, and what will happen after implementation. This assignment is due at the beginning of class on Friday September 6th.

Turn in:

- A memo to the boss (one page maximum) to explain what the problem is, what your plan is, and what you expect to happen after it is implemented.
- One page describing your overall model, including any assumptions you are making.
- Up to one page describing your model for the current situation.
- Up to one page describing your model for the predicted situation after implementing your recommendation.